

Capacitor Charging

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1 Introduction

When a capacitor C is charged through a series resistor R from a constant supply voltage V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right).$$

The product RC is called the time constant,

$$\tau = RC.$$

The time constant describes how quickly the capacitor charges.